ADV. MACHINE LEARNING PROJECT D2.1

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INTRODUCTION:

Our project is based around forecasting daily air traffic at an airport. We chose Palma de Mallorca Airport (LEPA) because it is one of the busiest in Spain and its traffic is almost entirely driven by tourism, which makes the seasonal patterns very visible in the data.

Air traffic is a classic example of a forecasting problem with strong seasonality and a clear set of factors that affect it (weather, holidays, etc.), which is why it fits this assignment well. It is also the type of problem that airports actually need to solve in real life, since they have to plan staff, slots and resources in advance.

FORECASTING OBJECTIVE & HORIZON:

Our goal is to predict the daily number of IFR (Instrument Flight Rules), including arrivals and departures, at Palma de Mallorca airport, fourteen days in advance.

We chose 14 days because it is the time horizon we believe would be useful for an airport. Predicting one or two days in advance is too easy, as most flights are already scheduled, and predicting more than a month in advance depends too heavily on factors like the economy or seasonal bookings, which aren't what we really want to model. Two weeks is a reasonable middle ground and aligns with the planning timelines airports use for staffing and operations.

STATIONARITY & RANDOM WALK ANALYSIS:

The series is not a random walk, and we can show this in a few ways.

If we plot the daily flights at Palma, the curve does not drift around freely like a random walk would. Instead it repeats itself every year: high in summer, low in winter, with the same shape over and over again. A random walk would not do this, since by definition it has no tendency to come back to any specific level.

To support this with proper tests, we will run the Augmented Dickey Fuller (ADF) test, which checks whether the series has a unit root (a unit root would mean it is a random walk). We expect to reject this hypothesis cleanly because of how strong the seasonality is. We will also use the KPSS test as a second check, since it has the opposite null hypothesis, and look at the autocorrelation function (ACF) of the differenced series, where we should see clear spikes at lag 7 (weekly) and lag 365 (yearly) instead of the flat ACF that a random walk would give.

The series is still non-stationary in the strict sense because of the yearly seasonality and the COVID-19 break in 2020, but that is a different problem from being a random walk and we will deal with it through seasonal differencing and a COVID indicator variable.

DATASET DESCRIPTION:

Our dataset comes from the Eurocontrol Aviation Intelligence Portal (https://www.ansperformance.eu/data/), which publishes daily IFR flight counts for every major European airport as a free CSV. We filter it to the code LEPA (Palma de Mallorca).

The data covers around ten years at daily frequency, from 2016 to late 2025, which gives us roughly 3,500 observations. That is more than enough for the models we want to compare.

There are two clear seasonal patterns we expect to find. The first is weekly: Fridays and Sundays are busier than the rest of the week, with a dip on Tuesdays. The second is yearly: traffic explodes between June and September because of summer tourism, and drops a lot in winter. On top of this, COVID-19 caused a huge drop in traffic in March and April 2020, and the recovery took about 18 months. We will treat the COVID period as a structural break rather than ignoring it.

As exogenous variables we plan to include weather data at Palma (temperature, rain, wind) from the Open-Meteo API, calendar features like the day of the week, the month and public holidays (we will also include German and UK school holidays since most tourists in Palma come from those countries), a COVID indicator variable, and the price of jet fuel as a slower macro variable. We are working with a single time series, although we might extend the project to a few other Spanish airports (Madrid, Barcelona, Bilbao, Málaga) if we have time, just to see how the same models do on different airport profiles.

PRIOR WORK / EXISTING FORECASTING MODELS:

Air traffic forecasting is one of the most studied problems in time series analysis, so there is a lot of previous work to look at. The classic starting point is the Box & Jenkins (1976) Airline Passengers dataset, which is the example used in almost every textbook to introduce SARIMA models.

Modern work uses mainly four kinds of models: classical statistical models like SARIMA and SARIMAX (still the standard baseline), Prophet from Meta (which handles multiple seasonalities and holidays well), deep learning models like LSTM and GRU, and gradient boosting (XGBoost, LightGBM) with hand-crafted lag and calendar features. There is also a more recent group of papers from 2021-2023 that focus specifically on how to deal with the COVID-19 break, which is something we will need to handle too.

Some of the studies we found that are closest to what we want to do are:

- **Maas (2021)** — makes 14-day-ahead forecasts of daily air traffic at a European hub during COVID, comparing ARIMA, SARIMA, SARIMAX and LSTM. SARIMAX with the flight schedule as exogenous variable was the best. This is the closest precedent to our project.
- **Hopfe, Lee & Yu (2023)** in the Journal of Air Transport Management — compares SARIMA against several neural networks (RNN, LSTM, GRU) at five US airports during and outside COVID. Neural networks beat SARIMA clearly during COVID but the gap almost disappears in normal times.
- **Li, Bi & Li (2017)** — applies SARIMA to a single airport (Kunming) and reports daily errors of 1-3%. We will use this as a reference point for our own SARIMA results.